Trust Flow-Driven Trusted Federated Learning in Edge Computing: A Study on Key Technologies

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Abstract. Federated learning (FL) supports collaborative edge intelligence without moving raw data, but open edge participation exposes aggregation to untrusted execution, model poisoning, stealthy backdoors, and severe Non-IID drift. Existing robust aggregation methods often tighten thresholds to reject malicious updates at the cost of removing legitimate long-tail clients. This paper proposes a trust-flow-driven trusted FL framework that couples hardware-anchored admission, runtime trust evidence, dual-stream reputation evolution, and risk-gated layered aggregation. The method separates long-term utility from short-term attack risk through a historical performance stream and an instant risk stream, then uses layer-wise gating, clipping, and survivor weight renormalization to suppress poisoned updates while preserving useful heterogeneous contributions. Experiments on Flower/PyTorch with CIFAR-10, 20 admitted clients, six mixed attack types, and five repeated runs show 92.31% accuracy and 10.21% ASR under a 30% malicious setting, with permanent-ban FPR close to zero.

Keywords: Federated Learning; Edge Computing; Trusted Execution Environment; Risk Modeling; Layered Aggregation; Backdoor Defense

1 Introduction

FL trains a global model through parameter exchange while keeping data on edge devices [1,25]. This design is well suited to edge intelligence, yet its aggregation phase becomes fragile when client identities, execution integrity, and update semantics cannot be fully trusted. Practical edge networks also exhibit device heterogeneity and strongly skewed data distributions, so robust aggregation has to distinguish malicious deviations from legitimate long-tail updates. This distinction becomes difficult under compound attacks, where adversaries may submit poisoned parameters, inject targeted backdoors, or exploit cross-round camouflage.

Prior work addresses different parts of this problem. Geometric and statistical defenses such as Krum and Trimmed Mean remove outlying updates [4,5], while FLTrust, FoolsGold, RaSA, and related detection schemes use clean references, similarity histories, or adaptive trust [6,7,11,12,14]. Backdoor-specific methods further examine trigger behavior or collusion patterns [2,3,15,16,17]. These mechanisms remain exposed to two recurring tensions: static software checks cannot guarantee that local execution was genuine, and aggressive filtering often mistakes Non-IID long-tail updates for attacks. TEE-based FL and trusted inference systems provide a hardware root of trust [18,19,20,21,22,23,24,13], but they usually do not by themselves solve semantic poisoning after a client has been admitted.

This paper proposes *Trust Flow*, a full-process trusted aggregation framework for edge FL. The main idea is to make trust evidence flow across admission, auditing, temporal state evolution, and aggregation rather than treating each round as an isolated filtering decision. The contributions are:

1. A TEE-anchored dynamic admission mechanism that combines attestation-style admission evidence with runtime behavioral entropy and Kalman filtering.
2. A dual-stream reputation model that decouples historical utility from instant attack risk, reducing the tension between malicious recall and permanent false bans.
3. A layer-wise risk-gated aggregation rule that applies differentiated admission thresholds, dynamic L2 clipping, and survivor weight re-normalization.
4. An end-to-end evaluation under Non-IID CIFAR-10 and mixed attacks, with additional stress tests and ablations.

2 Background and Related Work

Federated learning in edge computing. In the standard FL workflow, clients complete local training and submit parameter updates to the server, which aggregates them and redistributes the global model. This workflow keeps data local but makes the server heavily dependent on the authenticity and quality of uploaded updates. Edge computing further complicates this setting because clients differ in compute capability, communication stability, and data distribution. Clustered FL, adaptive pruning-expansion, and split federated learning have been proposed to address heterogeneity and resource constraints [8,9,10]. As the system becomes more open and layered, however, a larger attack surface appears around client admission, local execution, and update aggregation.

Poisoning and backdoor attacks. Attacks in FL mainly include untargeted poisoning and targeted backdoors. Untargeted poisoning, such as sign flipping and gradient scaling, aims to disrupt convergence. Targeted attacks, including label flipping, trigger-based backdoors, clean-label poisoning, and semantic backdoors, try to preserve main-task behavior while implanting targeted misclassification. Backdoor attacks are especially challenging because a malicious update can remain close to normal directions while shifting hidden features [2,3]. In a multi-round setting, attackers may also behave normally for several rounds to accumulate trust before injecting low-amplitude or intermittent malicious updates.

Active defenses and their limits. Robust aggregation methods typically rely on geometric distances, coordinate-wise statistics, clean anchors, or temporal similarity. Krum and Trimmed Mean reject or suppress outlying updates

[4,5]; FLTrust uses a server-side clean reference direction [6]; FoolsGold exploits historical similarity to reduce Sybil impact [7]; RaSA and recent detection frameworks further adapt aggregation or secure aggregation under edge-assisted settings [11,12,14]. Backdoor-oriented defenses, including FLPurifier, RoseAgg, and ShieldFL, improve protection against trigger or collusion patterns [15,16,17]. These defenses remain sensitive to strong Non-IID distributions: a strict threshold raises benign false positives, whereas a loose threshold lets stealthy attacks pass.

TEE-assisted trusted training. Trusted execution environments (TEEs), including Intel SGX and ARM TrustZone, provide isolated execution and protected key material. Prior studies show their value in verifiable inference, robust privacy-preserving FL, training integrity, attack mitigation, and industrial IoT trust sharing [18,19,20,21,22,23,24,13]. Nevertheless, TEE admission alone cannot decide whether a genuine training process produced semantically safe model updates. Trust Flow therefore combines hardware-rooted admission with cross-round risk modeling and layer-aware robust aggregation.

3 System Model and Threat Assumptions

The system contains a central server $\mathcal{S}$ and K edge clients $\mathcal{C} = \{c_1, \ldots, c_K\}$. In round t, admitted clients receive $W^{(t-1)}$, train locally, upload $\varDelta W_k^{(t)}$, and the server updates $W^{(t)} = W^{(t-1)} + \mathrm{Agg}(\{\varDelta W_k^{(t)}\})$. The global objective is

$$\min_W \mathcal{L}(W) = \sum_{k=1}^{K} p_k \mathcal{L}_k(W), \quad p_k = |\mathcal{D}_k| / \sum_j |\mathcal{D}_j|. \tag{1}$$

The edge setting follows a client-edge-cloud workflow. The client side provides local training and trust evidence, while the server audits updates, evolves trust states, and performs secure aggregation.

Threat model. The server is honest-but-curious and follows the aggregation protocol. Attackers may control admitted clients and perform label flipping, trigger backdoors, clean-label poisoning, semantic backdoors, sign flipping, and gradient scaling. This paper assumes an upper bound of $< 50\%$ malicious clients. The main experiments use 30% as a typical high-risk setting and additionally report a 50% boundary stress test. TEEs are assumed to protect attestation-critical code and keys against a compromised host OS, excluding physical probing and advanced microarchitectural side-channel attacks.

Design goals. Trust Flow aims to: (i) maintain convergence under mixed poisoning and backdoors; (ii) keep permanent-ban false positive rate (FPR) low for legitimate long-tail clients; and (iii) avoid heavyweight cryptographic overhead while preserving edge deployability. Figures 1 and 2 summarize the physical workflow and the threat surface.

The central difficulty is that these goals are coupled. In a pure filtering strategy, strengthening the rejection rule improves malicious recall but also removes benign clients whose local distributions deviate from the majority. Conversely,

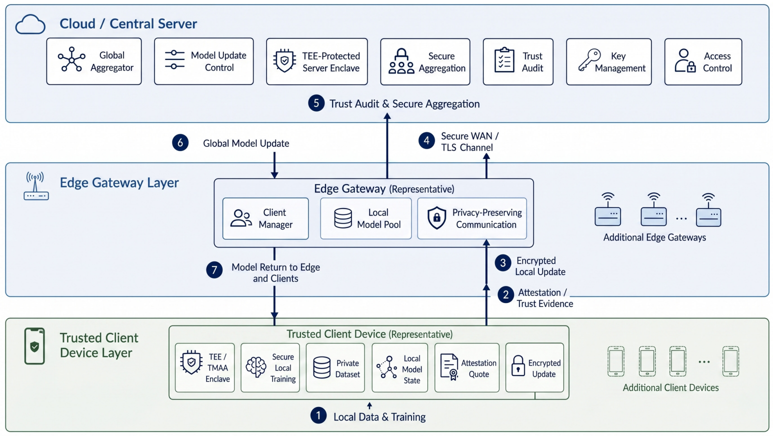


Fig. 1. TEE-anchored client-edge-cloud architecture.

relaxing the rule protects long-tail clients but allows low-amplitude poisoning to persist. Trust Flow treats this as a state-separation problem rather than a threshold-tuning problem: admission evidence, content quality, historical contribution, and instant risk remain visible as different signals until the final layer-wise aggregation decision.

4 Trust-Flow Framework

Trust Flow divides each FL round into four linked stages: admission and runtime trust assessment, reference-based content auditing, dual-stream state evolution, and risk-gated layered aggregation. Figure 3 shows the full defense loop.

At a higher level, the framework is a trust-state transfer pipeline. Phase one outputs $TrustScore$, which captures whether the client has a trustworthy basis for participation. Phase two outputs $ContentScore$, which measures whether the current update agrees with the clean reference and the peer group. Phase three deposits these signals into $HistPerf$ and $RiskEMA$, separating long-term contribution from short-term abnormality. Phase four fuses the states into $RawScore$ and performs differentiated layer-wise aggregation. This design avoids treating a low single-round score as immediate evidence of maliciousness, which is important for long-tail clients under strong Non-IID data.

4.1 Admission and Content Auditing

Each admitted client must pass a TEE-style admission check. Let $M_{\text{attest},k} \in \{0, 1\}$ denote whether the attestation report is valid. During initialization, the client invokes a device-unique key and generates a remote attestation quote that includes integrity measurements such as PCR chains, code segment signatures, and runtime configuration. Only clients passing this check receive communication privileges. During local training, the trusted management agent monitors

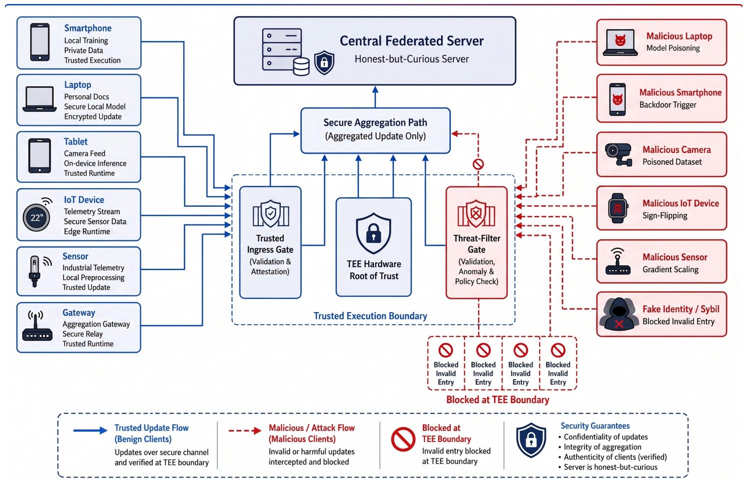


Fig. 2. Composite poisoning and penetration threat model.

behavioral sequences $\mathcal{X}_k = \{\Delta\text{grad}, \Delta\text{loss}, \text{instr\_dist}\}$ and maps their entropy to measurement uncertainty:

$$H(\mathcal{X}_k) = -\sum_i p(x_i)\log p(x_i), \qquad R_k^{(t)} = \exp(\lambda H(\mathcal{X}_k^{(t)})). \tag{2}$$

Following the information-theoretic filtering view [26] and multidimensional trust evaluation [27], the trust estimate is updated by

$$\hat{T}_k^{(t)} = \hat{T}_k^{(t-1)} + K_k^{(t)}(Z_k^{(t)} - \hat{T}_k^{(t-1)}), \quad TrustScore_k = \hat{T}_k^{(t)}(1 + P_k^{(t)})^{-1}. \tag{3}$$

High-entropy runtime behavior increases $R_k^{(t)}$, reducing the Kalman gain and preventing abrupt anomalous observations from being absorbed into the long-term trust estimate. The covariance $P_k^{(t)}$ serves as a confidence boundary: smaller covariance indicates more reliable trust estimation. Thus the admission result is not a one-time binary decision, but a continuously updated privilege factor that can be carried into later auditing and aggregation.

The server then constructs a clean guiding direction. If a small clean validation gradient is available, $g_{root} = g_{root\_clean}$; otherwise, it uses high-trust and low-risk clients:

$$\omega_k^{ref} = TrustScore_k(1 - RiskEMA_k^{prev})^2, \quad g_{root} = \sum_k \frac{\omega_k^{ref}}{\sum_j \omega_j^{ref}} \Delta W_k. \tag{4}$$

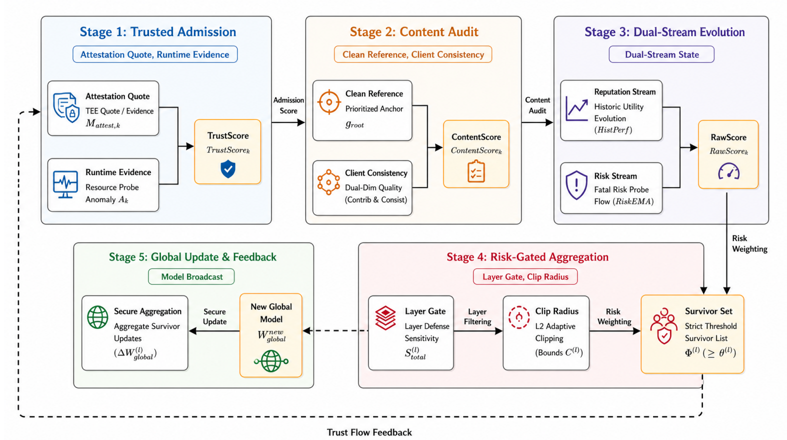


Fig. 3. Overview of the five-phase trust-flow defense loop.

For each client update, content quality combines reference consistency and peer consistency:

$$S_{contrib,k} = \max(0, \alpha_1 + \alpha_2 \cos(\Delta W_k, g_{root})), \tag{5}$$

$$S_{consist,k} = \frac{\sum_{j \neq k} TrustScore_j \max(0, \alpha_1 + \alpha_2 \cos(\Delta W_k, \Delta W_j))}{\sum_{j \neq k} TrustScore_j + \varepsilon}, \tag{6}$$

$$ContentScore_k = \frac{(1 + \beta^2_{fusion}) S_{consist,k} S_{contrib,k}}{\beta^2_{fusion} S_{consist,k} + S_{contrib,k} + \varepsilon}. \tag{7}$$

Here $\beta_{fusion} > 1$ increases the influence of the clean direction and helps resist collusive drift.

4.2 Dual-Stream State Evolution

Single reputation scores confuse low utility caused by data skew with high risk caused by attacks. Trust Flow therefore separates client state into a historical utility stream and an instant risk stream. The historical stream uses a Beta reputation update:

$$r^{good}_k = \text{clip}\left((ContentScore_k - \mu_{content})/(\sigma_{content} + \varepsilon), 0, 1\right), \tag{8}$$

$$\alpha^{(t)}_k = \lambda_h \alpha^{(t-1)}_k + r^{good}_k, \quad \beta^{(t)}_k = \lambda_h \beta^{(t-1)}_k + (1 - r^{good}_k), \tag{9}$$

$$HistPerf^{(t)}_k = \alpha^{(t)}_k / (\alpha^{(t)}_k + \beta^{(t)}_k + \varepsilon). \tag{10}$$

This channel is intentionally slow: a legitimate long-tail node can recover after temporary low content scores. A low $HistPerf$ means the node has recently contributed little; it leads to soft isolation rather than permanent removal. The

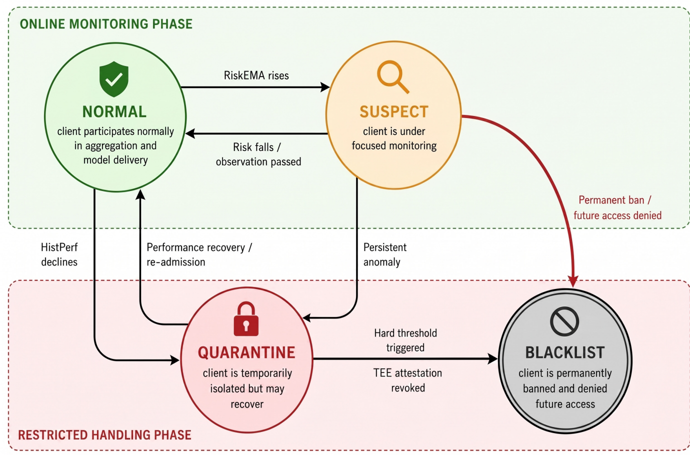


Fig. 4. Client state-transition automaton driven by historical utility and instant risk.

risk stream is fast and uses a max-probe rule:

$$Risk_k^{inst} = \max\{r_{report}, r_{grad}, r_{probe}, r_{pixel}, r_{trigger}, r_{sign}, r_{peer}\}, \tag{11}$$

$$RiskEMA_k^{(t)} = \beta_r RiskEMA_k^{(t-1)} + (1-\beta_r) Risk_k^{inst}. \tag{12}$$

The probes include gradient direction/amplitude, small clean-set loss increase, trigger activation drift, report anomalies, sign inconsistency, and peer disagreement. These two streams meet only when computing aggregation privilege:

$$\begin{aligned} RawScore_k = {} & (TrustScore_k)^{\alpha}(ContentScore_k)^{\beta}(HistPerf_k^{(t-1)})^{\gamma} \\ & \times (1 - RiskEMA_k^{(t-1)})^{p}. \end{aligned} \tag{13}$$

The resulting state machine distinguishes NORMAL, SUSPECT, QUARANTINE, and BLACKLIST states, as shown in Fig. 4. A sleeper-burst attacker may accumulate high $HistPerf$ during early benign rounds, but a later trigger injection activates $r_{trigger}$ or r_{probe} and increases $RiskEMA$, blocking the attack even when historical utility is high. Conversely, a long-tail benign client with low current similarity is mainly handled through the slow utility stream and can recover after later positive contributions.

4.3 Risk-Gated Layered Aggregation

Backdoors often concentrate in deeper layers, whereas shallow feature layers may carry useful heterogeneous information. Trust Flow therefore assigns each layer l a sensitivity score:

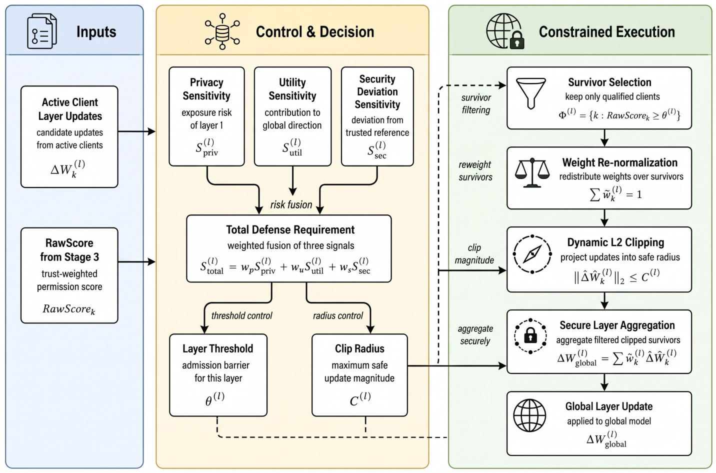


Fig. 5. Risk-gated layer-wise clipping and differentiated aggregation.

$$S_{privacy}^{(l)} = \exp(-\tau_p l/L), \quad S_{utility}^{(l)} = \|g_{ref}^{(l)}\|_2/(\max_m \|g_{ref}^{(m)}\|_2 + \varepsilon), \tag{14}$$

$$S_{security}^{(l)} = 1 - \frac{\sum_{k\in\mathcal{A}} TrustScore_k \cos(\Delta W_k^{(l)}, g_{ref}^{(l)})}{\sum_{k\in\mathcal{A}} TrustScore_k + \varepsilon}, \tag{15}$$

$$S_{total}^{(l)} = w_p S_{privacy}^{(l)} + w_u S_{utility}^{(l)} + w_s S_{security}^{(l)}. \tag{16}$$

The layer threshold and clipping radius are

$$\theta^{(l)} = \mu_{base} + \lambda_s S_{total}^{(l)}, \qquad C^{(l)} = C_{base}/(S_{total}^{(l)} + \varepsilon_c). \tag{17}$$

Survivors are $\Phi^{(l)} = \{k \in \mathcal{A} \mid RawScore_k \geq \theta^{(l)}\}$, and their weights are renormalized:

$$\tilde{w}_k^{(l)} = \frac{RawScore_k}{\sum_{j\in\Phi^{(l)}} RawScore_j + \varepsilon}, \quad \widehat{\Delta W}_k^{(l)} = \frac{\Delta W_k^{(l)}}{\max(1, \|\Delta W_k^{(l)}\|_2/C^{(l)})}. \tag{18}$$

The final update is $\Delta W_{global}^{(l)} = \sum_{k\in\Phi^{(l)}} \tilde{w}_k^{(l)} \widehat{\Delta W}_k^{(l)}$. Weight re-normalization prevents the removal of risky clients from shrinking the effective learning rate.

This layer-wise control is intended to preserve useful shallow representations while hardening layers that show stronger attack sensitivity. If the security term of a deep layer rises, the layer threshold increases and the clipping radius decreases, making it harder for low-trust or high-amplitude updates to affect that layer. If the shallow layers remain close to the reference direction, they keep more permissive thresholds. This differentiated treatment is important because a uniform clipping radius can over-suppress benign feature extraction layers and slow convergence, especially after malicious clients have already been removed from the survivor set.

4.4 Analytical View

The dual-stream structure gives a simple way to interpret the FPR/ASR trade-off. Suppose the single-round content score of a legitimate long-tail client follows a perturbed distribution with expectation μ_{clean} and variance σ^2_{clean}. A rigid threshold can eliminate this client whenever its current score is below the threshold, even if the deviation comes from Non-IID data. After exponential smoothing, the steady-state variance of the historical utility estimate follows the standard low-pass form:

$$\mathrm{Var}(HistPerf_k^{(\infty)}) = \frac{1-\beta_h}{1+\beta_h}\sigma^2_{clean}. \tag{19}$$

As β_h approaches 1, short-term benign fluctuations are suppressed and the client is less likely to face sustained removal. In contrast, the instant risk stream responds to attack impulses. If a backdoor round produces an abnormal probe value, the max operator prevents that event from being averaged away before entering the EMA. Processing long-term contribution and short-term risk in separate spaces therefore reduces the single-score conflict: long-tail updates are smoothed rather than banned immediately, while abrupt malicious signals retain enough strength to trigger isolation.

4.5 Risk Probe Computation

The instant risk stream uses three central probes. The gradient probe measures direction and amplitude deviation:

$$r^{(t)}_{grad,k} = \lambda_d \left(1 - \frac{\Delta W_k^{(t)} \cdot g^{(t)}_{root}}{\|\Delta W_k^{(t)}\| \|g^{(t)}_{root}\|}\right) + \lambda_m \max\left(0, \frac{\|\Delta W_k^{(t)}\|_2 - \mu^{(t)}}{\sigma^{(t)}}\right). \tag{20}$$

The clean probe checks whether applying a client update increases cross-entropy on a small server-side clean set $\mathcal{D}_{probe}$:

$$r^{(t)}_{probe,k} = \mathrm{ReLU}\left(\mathcal{L}_{CE}(W^{(t-1)} + \Delta W_k^{(t)}; \mathcal{D}_{probe}) - \mathcal{L}_{CE}(W^{(t-1)}; \mathcal{D}_{probe})\right). \tag{21}$$

The trigger probe compares high-level feature activations by KL divergence, $r^{(t)}_{trigger,k} = \mathrm{KL}(\mathcal{H}(A_{base}) \| \mathcal{H}(A_k))$, which helps identify clean-label or semantic backdoors whose low-level gradient directions remain close to benign updates.

5 Experiments

5.1 Setup

Experiments use Flower 1.5.0 and PyTorch on CIFAR-10 with a lightweight CNN. Simulations are deployed on an Inspur server with dual CPUs and five NVIDIA A10 GPUs. The software stack is Ubuntu 20.04.6 LTS and CUDA

Table 1. Compact experimental configuration and main hyperparameters.

Category	Setting	Value
FL task	Dataset/model/admitted clients	CIFAR-10/light CNN/$K = 20$
Optimization	Local epochs, lr, momentum, rounds	3, 0.001, 0.9, 30
Data heterogeneity	Shared pool, Group A, Group B/C	5000, $\alpha = 1.0$, $\alpha = 0.1$
Trust streams	$\alpha_1, \alpha_2, \beta_{fusion}$; β_h, β_r	$0.6, 0.4, 2.0$; $0.9, 0.85$
Layer gating	fusion exponents; μ_{base}, λ_s	$3.0, 1.0, 0.5, 1.0$; $0.4, 1.2$

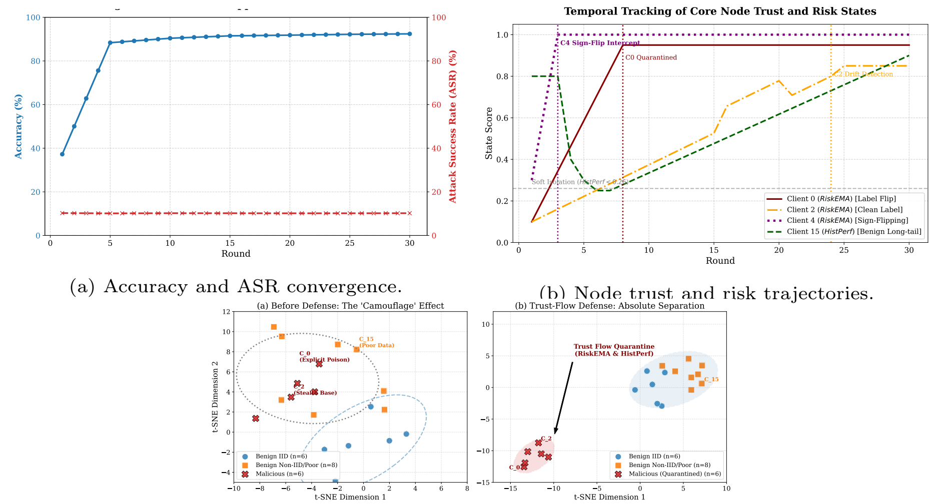
12.8. To support reproducibility, complete logs, source code, and startup scripts are preserved. The 50,000 training images are split among 20 admitted clients. A shared 5,000-sample balanced pool supplies a common alignment basis; clients 6–11 follow Dirichlet $\alpha = 1.0$, and clients 12–19 use strong long-tail partitions with $\alpha = 0.1$.

The formal 30% attack setting uses six malicious clients: Client 0 performs label flipping with $\gamma = 0.5$; Client 1 injects a trigger backdoor into 20% of samples and targets class 0; Client 2 uses clean-label backdoor perturbations; Client 3 uses semantic backdoor bias; Client 4 applies sign flipping; and Client 5 amplifies updates by a factor of 100. Five Sybil nodes without valid certificates and three free-rider nodes with forged low workload are used to validate Phase-One gating. These eight nodes are intercepted before the statistical analysis, so the later metrics are computed over the 20 admitted clients. Each run uses 30 rounds and three local epochs. Aggregate Acc/ASR values are averaged over five seeds; state trajectories are shown from one representative seed.

Accuracy and ASR are reported as mean and standard deviation over five independent runs. Permanent-ban FPR is computed over benign clients that are irreversibly blacklisted, while temporary SUSPECT or QUARANTINE states are treated as recoverable control actions. The single-run trajectory figures are taken from a fixed seed to make node-level transitions readable; aggregate tables use the repeated runs. All baselines are executed under the same node partitioning, attack proportion, optimizer configuration, communication rounds, and Non-IID split.

5.2 Overall Defense Performance

Figure 6 summarizes convergence, ASR, trust-state trajectories, and feature separation. Under 30% mixed attacks, Trust Flow converges to 92.31% accuracy and reduces ASR to 10.21%, close to the random-guess baseline for CIFAR-10. Client-state tracing shows that explicit poisoning and parameter attacks are blocked early, while stealthier clean-label drift is captured after persistent probe evidence. Legitimate long-tail nodes may enter SUSPECT or QUARANTINE temporarily, but their risk stream does not accumulate enough evidence for permanent banning.



(a) Accuracy and ASR convergence.

(b) Node trust and risk trajectories.

(c) t-SNE separation before/after defense.

Fig. 6. Main defense behavior under mixed attacks.

The node trajectories expose different decision paths. Label flipping and trigger backdoor clients show sustained high r_{grad} and r_{probe} values and enter the blacklist by round 8. Semantic backdoor behavior first causes soft isolation and then removal after persistent risk accumulation. The clean-label attacker is harder to detect because its local loss appears normal; it is identified after the drift-combination rule remains active for several rounds. Sign flipping sharply reduces the directional contribution score, whereas gradient scaling is caught by amplitude and layer-wise clipping. In contrast, the long-tail legitimate client may show low content quality early, but its risk stream does not keep rising, allowing it to recover through later $HistPerf$ updates.

The t-SNE visualization provides a complementary view. Before trust-flow intervention, malicious and legitimate long-tail clients can occupy overlapping regions because both may deviate from the majority direction. After dual-stream supervision, malicious clients form separable clusters associated with high-risk probes, whereas legitimate long-tail clients remain connected to the main aggregation region. This supports the design claim that historical utility and instant risk should not be collapsed into one score too early.

Compared with FedAvg, robust aggregation baselines reduce ASR but either lose substantial accuracy or remove many benign long-tail clients. Krum lowers ASR to 12.15% but its distance rule excludes many legitimate long-tail updates, leading to 64.20% accuracy. Trimmed Mean is more stable than Krum in some rounds, yet coordinate-wise truncation cannot suppress targeted backdoor behavior sufficiently. FLTrust benefits from a clean anchor but remains vulnerable when legitimate Non-IID updates deviate from the anchor. Trust Flow preserves benign heterogeneous contributions through historical smoothing, while the instant risk stream prevents high-reputation attackers from exploiting past behav-

Table 2. Performance comparison under 30% mixed attacks and boundary stress testing.

Method/Setting	Core mechanism	Acc.	ASR	Perm. FPR
FedAvg [25]	no defense	86.15 ± 1.20	92.34 ± 5.10	N/A
Krum [4]	Euclidean selection	64.20 ± 2.80	12.15 ± 0.90	64.20
Trimmed Mean [5]	coordinate trimming	71.35 ± 2.40	38.60 ± 4.20	N/A
FLTrust [6]	clean anchor weighting	81.25 ± 1.50	15.42 ± 1.10	21.40
Trust Flow	dual stream + layered gating	$\mathbf{92.31 \pm 0.09}$	$\mathbf{10.21 \pm 0.05}$	**0.00**
Boundary stress: 50% malicious	10/20 malicious	91.81 ± 0.10	10.46 ± 0.06	0.00

FPR denotes irreversible false bans of benign clients. Temporary SUSPECT/QUARANTINE states are recoverable and reported separately through trajectories.

Table 3. Ablation and overhead summary.

Study	Variant / architecture	FPR or overhead	TPR / time
Reputation	Single-stream EMA, aggressive	18.25% FPR	95.12% TPR
Reputation	Single-stream EMA, conservative	2.10% FPR	48.33% TPR
Reputation	HistPerf-only	0.15% FPR	21.05% TPR
Reputation	**Dual-stream Trust Flow**	**0.05% FPR**	**98.50% TPR**
Overhead	Standard FedAvg	0 KB extra	2.10 s aggregation
Overhead	Trust Flow	~15 KB TrustReport	4.85 s incl. audit

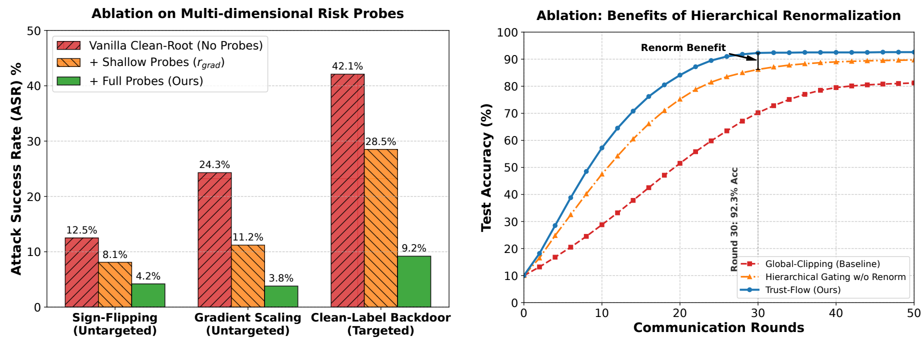
ior. The 50% result shows that the framework remains stable near the theoretical boundary, with 91.81% accuracy and 10.46% ASR.

5.3 Ablation, Sensitivity, and Overhead

Table 3 shows that a single-stream reputation model suffers from an FPR/TPR tradeoff: aggressive thresholds increase false bans, while conservative thresholds miss sleeper attacks. HistPerf-only protects benign nodes but lacks short-term attack sensitivity. The dual-stream design reaches high recall while keeping permanent false bans low. The overhead is moderate: TrustReports add about 15 KB per upload, and server aggregation time increases from 2.10 s to 4.85 s per round.

Figures 7 and 8 further decompose the effect of risk probes, weight renormalization, and data heterogeneity. Clean reference plus shallow probes handles sign flipping and gradient scaling, but clean-label backdoors remain difficult when gradient directions are collinear with benign updates. Adding cross-entropy and high-level activation probes reduces clean-label ASR below 9.2%. Re-normalization also matters: layer-wise gating without survivor weight correction creates an implicit learning-rate decay, while full Trust Flow converges within the 30-round budget. Under Dirichlet $\alpha = 0.1$, Krum and FLTrust degrade sharply, whereas Trust Flow remains near 92% accuracy.

The sensitivity curve confirms that traditional robust aggregation behaves well when data are close to IID, but deteriorates as α decreases. With $\alpha = 100$ or $\alpha = 1.0$, Krum and FLTrust remain close to the proposed method. In the long-tail setting $\alpha = 0.1$, however, their filtering rules increasingly treat legitimate



(a) Multi-dimensional risk-probe ablation.

(b) Weight re-normalization effect.

Fig. 7. Component-level ablation of risk probes and re-normalization.

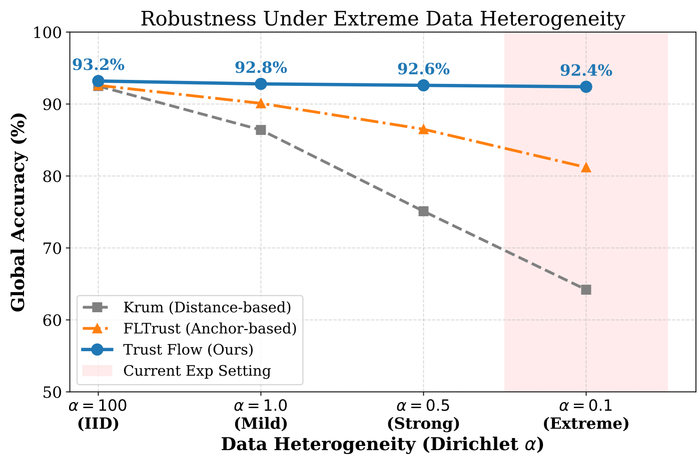


Fig. 8. Robustness stress test under varying data heterogeneity.

drift as malicious deviation. Trust Flow is less sensitive to this shift because content evidence is not used alone; it is integrated with historical utility and instant risk. The fusion parameter β_{fusion} controls how strongly the clean reference dominates peer consistency. A smaller value retains more long-tail clients but raises ASR, while a larger value improves attack suppression but can over-penalize heterogeneous updates. The default $\beta_{fusion} = 2.0$ provides the best observed Acc/ASR balance.

The overhead analysis indicates that Trust Flow mainly shifts cost to the server. The edge side appends a TrustReport of roughly 15 KB and performs lightweight runtime monitoring. The server performs reference construction, pairwise consistency auditing, dual-stream updates, and layer-wise clipping, increasing aggregation time from 2.10 s to 4.85 s in the tested setup. Because wide-area edge FL rounds usually spend far more time on communication and synchronization than on aggregation arithmetic, this increase remains acceptable for the target scenario.

6 Discussion and Conclusion

Trust Flow integrates admission evidence, content auditing, temporal reputation, and layer-wise aggregation into a single trust-state pipeline. Its main benefit is not only stronger attack suppression, but a cleaner separation between two causes of low update similarity: benign long-tail data and malicious manipulation. This distinction allows the method to keep permanent-ban FPR near zero while maintaining high recall against mixed attacks.

Several limitations remain. Because publicly verifiable implementations are unavailable for several emerging TEE-FL methods, this study does not include them as direct baselines under the same experimental setting. The current probe design is also tailored to vision classification, especially CIFAR-10 backdoor behavior. Future work will release the source code and experimental scripts with the final manuscript to support reproduction, evaluate larger edge deployments, and extend the multidimensional risk probes to NLP and federated fine-tuning scenarios.

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